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$$\begin{aligned}\int \frac{x^3 + 1}{x - 1} dx &= \int \frac{x^3 - 1 + 2}{x - 1} dx = \int \frac{(x - 1)(x^2 + x + 1)}{x - 1} dx + \int \frac{2}{x - 1} dx \\ &= \int (x^2 + x + 1) dx + \int \frac{2}{x - 1} dx = \int x^2 dx + \int x dx + \int 1 dx + \int \frac{2}{x - 1} dx\end{aligned}$$

$$u = x - 1 \Rightarrow du = dx$$

$$\int \frac{2}{x - 1} dx = \int \frac{2}{u} du = 2 \ln |u| + c = 2 \ln |x - 1| + c$$

$$\int \frac{x^3 + 1}{x - 1} dx = \frac{x^3}{3} + \frac{x^2}{2} + x + 2 \ln |x - 1| + c$$

References